

BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: E Batch: 3 Practical No: 1

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Subject Name: Deep Learning

Problem Statement: Install and configure TensorFlow/Keras in Google Colab. Perform data preprocessing, normalization, train-test split, and visualization on a sample dataset

Algorithm:

- 1) Open Google Collab.
- 2) Import the required Python libraries: TensorFlow, Keras, NumPy and Matplotlib
- 3) Verify that TensorFlow is installed by checking its version.
- 4) Load the MNIST handwritten digit dataset from Keras.
- 5) Display the shape of the training and testing datasets.
- 6) Visualize a few sample images from the dataset.
- 7) Normalize the pixel values by dividing each pixel by 255 so that values lie between 0 and 1.
- 8) Verify the normalization process.
- 9) Display normalized images for comparison.
- 10) Print the final dataset dimensions.
- 11) The dataset is now preprocessed and ready for deep learning model training.

Code:

```
import tensorflow as tf

from tensorflow import keras

import numpy as np

import matplotlib.pyplot as plt

print("=" * 50)
```

```
print("TensorFlow Version:", tf.__version__)

print("=" * 50)

(x_train, y_train), (x_test, y_test) = keras.datasets.mnist.load_data()

print("\nDataset Information")

print("-" * 50)

print("Training Images Shape :", x_train.shape)

print("Training Labels Shape :", y_train.shape)

print("Testing Images Shape :", x_test.shape)

print("Testing Labels Shape :", y_test.shape)

print("\nDisplaying Sample Images Before Normalization...")

plt.figure(figsize=(10,5))

for i in range(10):

    plt.subplot(2,5,i+1)

    plt.imshow(x_train[i], cmap='gray')

    plt.title("Label: {}".format(y_train[i]))

    plt.axis('off')

plt.tight_layout()

plt.show()
```

```
x_train = x_train.astype('float32') / 255.0

x_test = x_test.astype('float32') / 255.0


print("\nAfter Normalization")

print("-" * 50)

print("Minimum Pixel Value :", x_train.min())

print("Maximum Pixel Value :", x_train.max())


print("\nDisplaying Images After Normalization...")


plt.figure(figsize=(10,5))


for i in range(10):

    plt.subplot(2,5,i+1)

    plt.imshow(x_train[i], cmap='gray')

    plt.title("Label: {}".format(y_train[i]))

    plt.axis('off')


plt.tight_layout()

plt.show()


print("\nFinal Dataset Summary")

print("-" * 50)
```

```

print("Training Images :", x_train.shape)

print("Training Labels :", y_train.shape)

print("Testing Images :", x_test.shape)

print("Testing Labels :", y_test.shape)


print("\nTraining Pixel Range :", x_train.min(), "to", x_train.max())

print("Testing Pixel Range :", x_test.min(), "to", x_test.max())


print("\nPractical Completed Successfully!")

```

Output:

